Physics-based Predictive Modelling to Decipher Phase Transitions and Gas Evolution in High-energy NMC811 Electrodes

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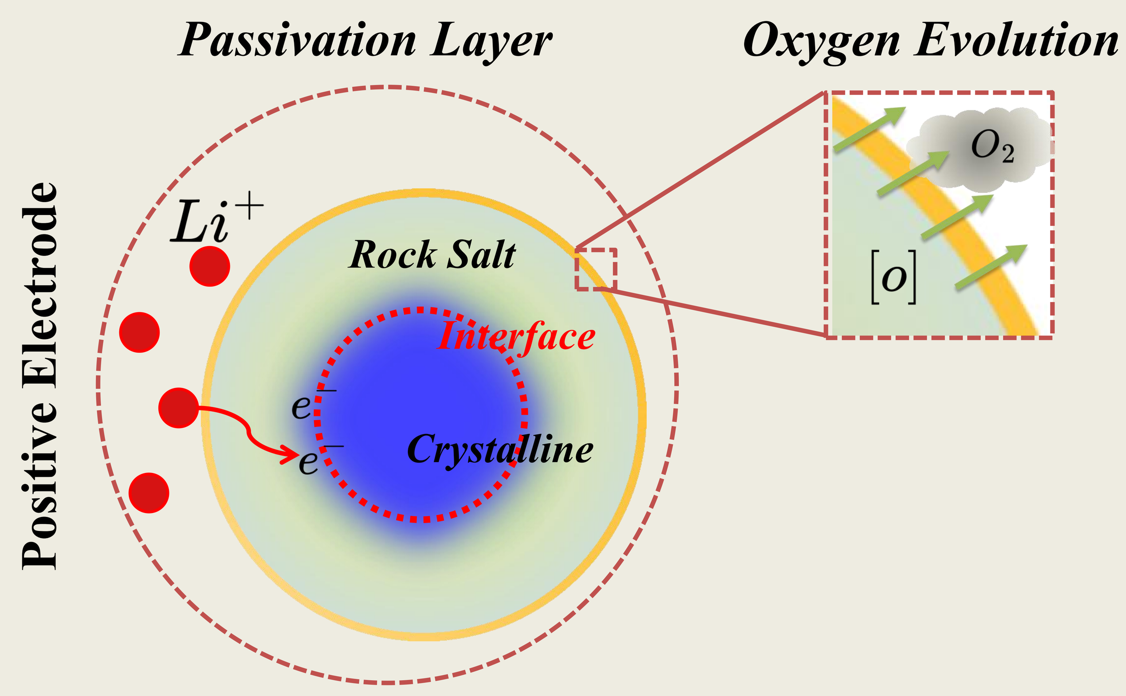
Overview

Introduction

- Novelty: Thermally-Coupled @ CC-CV
- Utility: degradation-Integrated
- Background: Single Particle Model

Problem Statement:-
Ni-rich layered cathodes (eg., NMC811) offer high specific capacities but suffer from severe degradation at high voltage due to **oxygen evolution**, **structural reconstruction**, and **passivation layer growth**. These degradation mechanisms induce **loss of active material (LAM)**, **loss of lithium inventory (LLI)**, and **thermal instability**, leading to capacity and power fade over repeated cycling.

Degradations



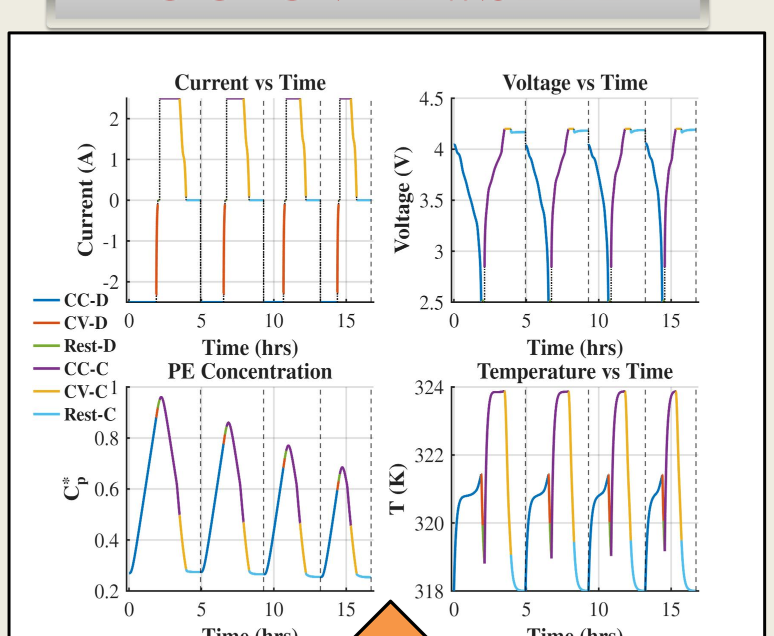
References

❑ **Publications**

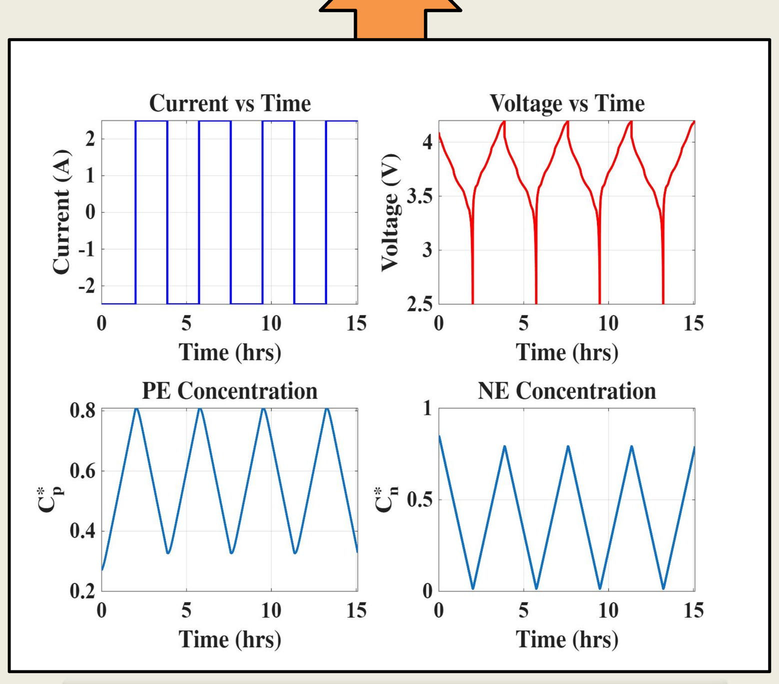
[1] **Ghosh, A.**, et al. J. Electrochem. Soc., **2021**, 168, 020509.
[2] Chen, C., et al. J. Electrochem. Soc., **2020**, 167, 080534.
[3] Guo, M., et al. J. Electrochem.Soc., **2011**, 158, A122.
[4] Zhuo, M., et al. J. Power Sources. **2023**, 556, 232461.

Results

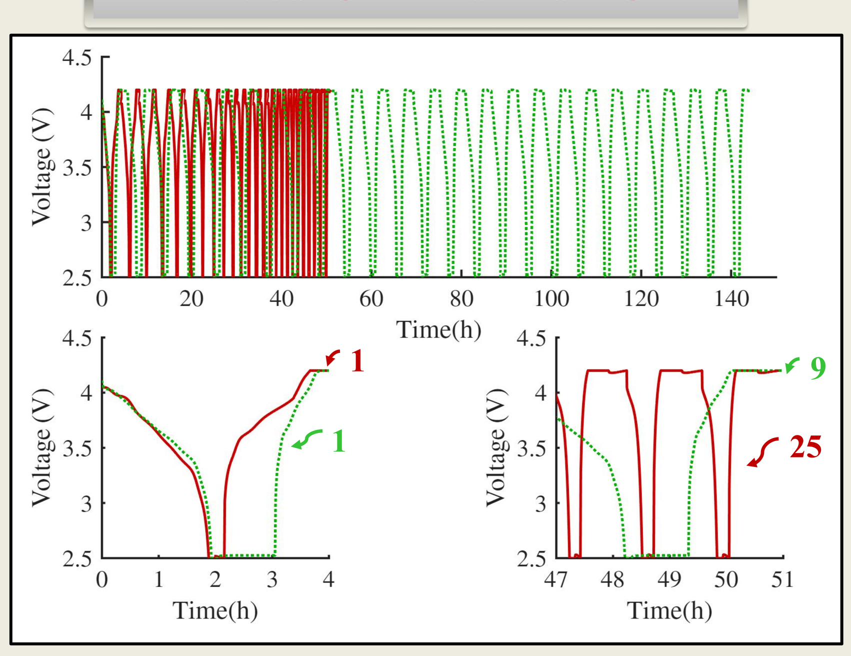
CC-CV-T-*d*SPM



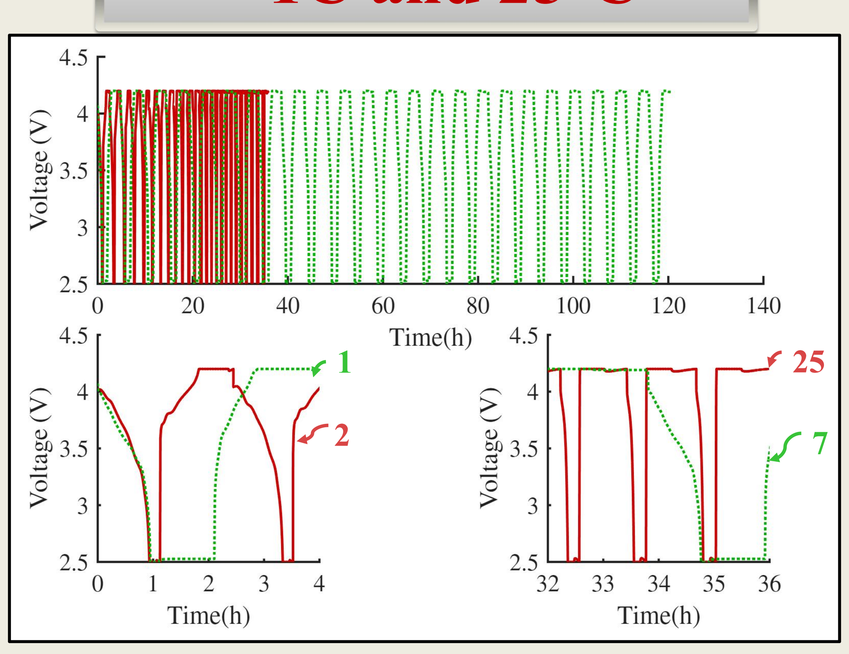
CC SPM



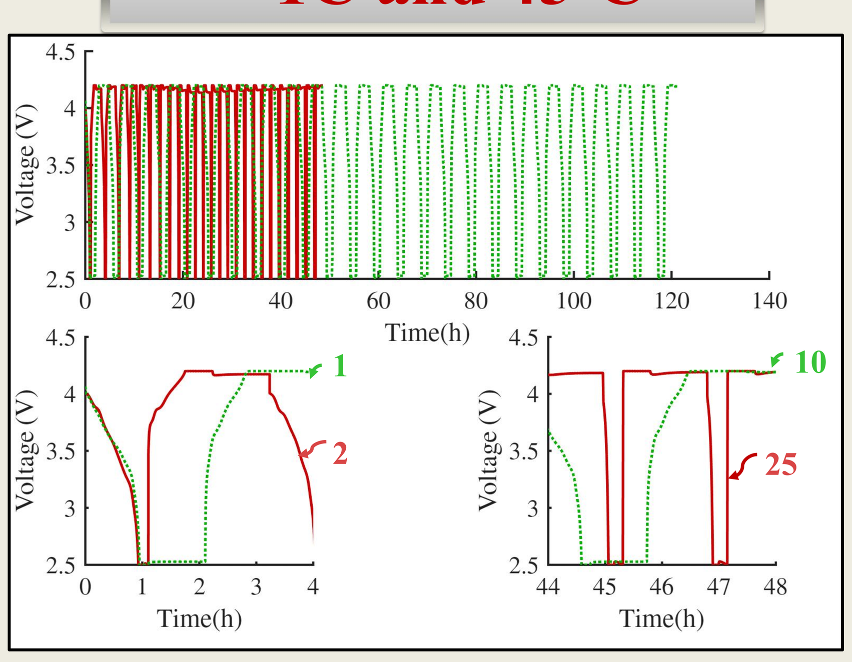
0.5C and 25°C



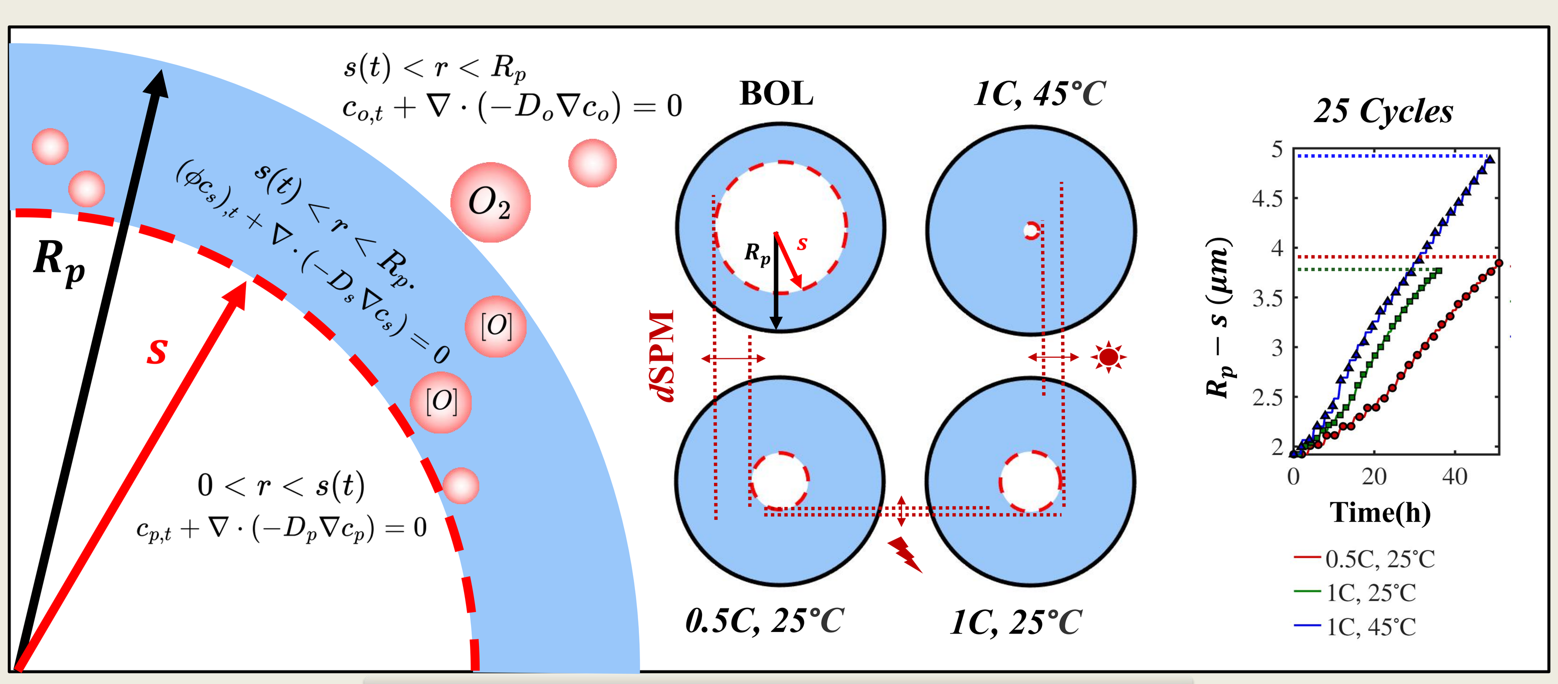
1C and 25°C



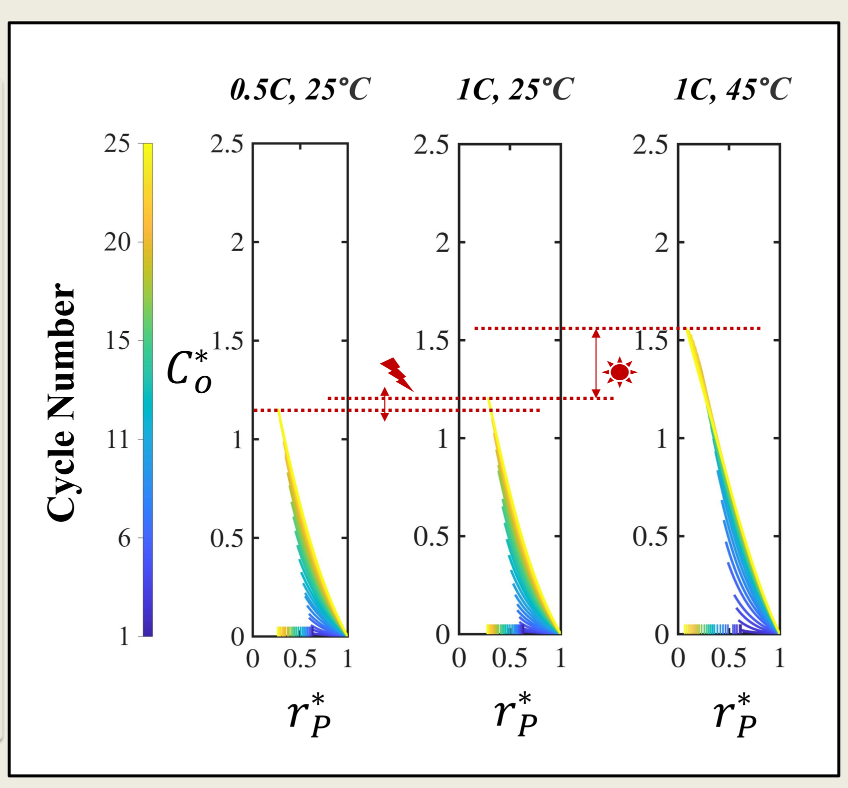
1C and 45°C



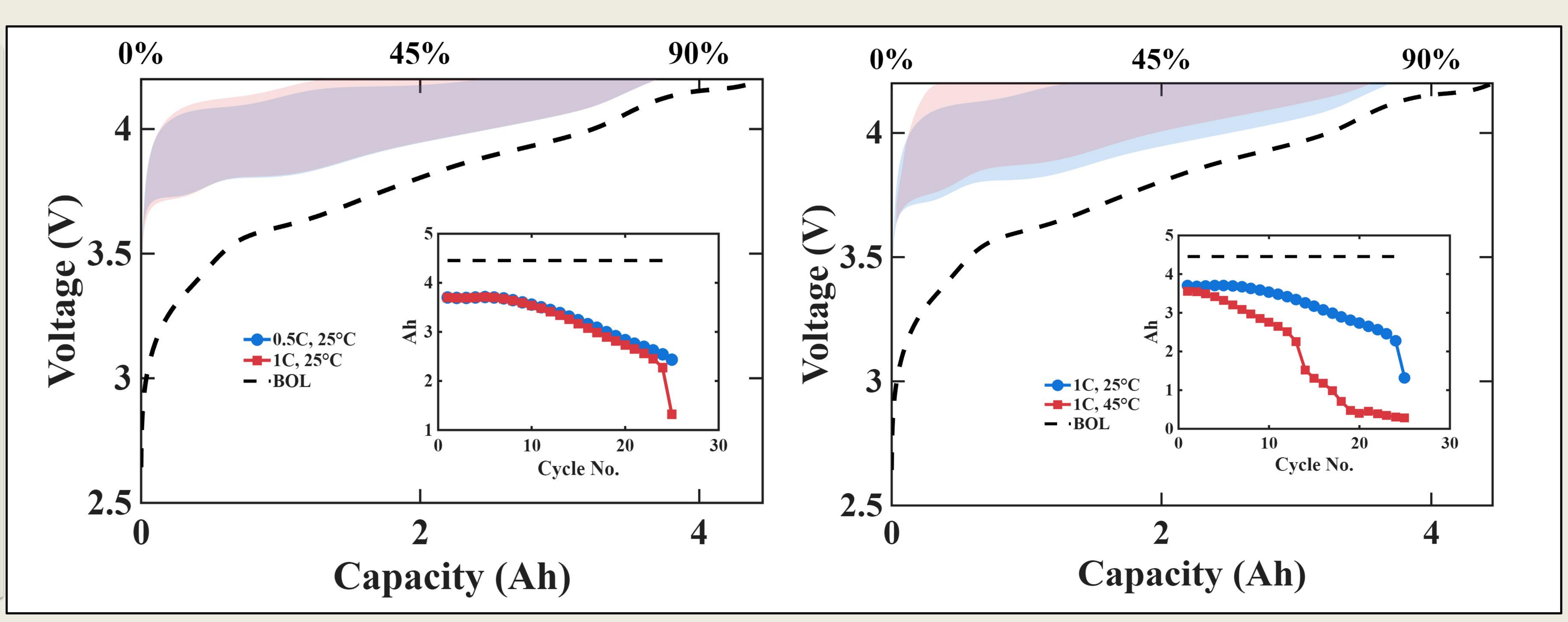
Shrinking Core



O_2 Evolution



0.5C and 1C at 25°C / 1C at 25°C and 45°C



Conclusion

INITIAL STABLE OPERATION

NMC Crystalline	Balanced Li^+ Transport	Intact Electrode Structure

ONSET OF DEGRADATION

Loss of Active Material	Passivation Growth	Loss of Lithium Inventory

ACCELERATED VOLTAGE DROP/ FAILURE

Severe Impedance Growth	Structural Breakdown	Thermal Instability

Thanks